

Five Hundred Dollars and Four Hours

Design the gate for a generator with no references and a real budget

Duration: 45 to 60 minutes

Group size: 3

Output: One triaged slate, one costed gate, and a signed budget line

Module context: Model Evaluation Metrics and Experiment Tracking, at the end. Everything from today is available: surface metrics, semantic metrics, judged evaluation, bias controls, calibration, reference-free evaluation, and significance testing. Nothing is held back. That is the difficulty.

THE PREMISE You are shipping the Cordwell catalog blurb generator. There are no reference blurbs, there never will be, and the entire left half of today's material is therefore unavailable to you. You have an attribute table, a style guide, and 40,000 historical blurbs. You have 500 dollars of judge calls per release and four hours of human review. Design the gate.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

You have every tool from today and not enough money to use them all. That is the whole exercise. A gate that ignores the budget is not a design, it is a wish list.

THERE ARE HINTS AND USING THEM IS NOT CHEATING Section 12 has a stuck ladder for the five hardest questions, three levels each. Take one level at a time and stop as soon as you are moving again. Reading all three levels for a question you were about to solve is the only way to waste them.

2. Your role

Each person takes one role. Each owns a different definition of the word shippable, and the friction between those definitions is the discussion.

Role	You own	The question you keep asking
Catalog merchandising lead	Every SKU your gate blocks from publishing, and the revenue on a page that stays empty	How many good blurbs does this gate throw away, and who unblocks them?
Evaluation engineer	The gate, the judge, and the 500 dollars	What does this cost per release, and what does it buy that the cheaper thing does not?
Brand and compliance partner	The style guide, and every claim that creates legal exposure	Which of these failures ends up in front of a regulator, and would we know?

3. What you are shipping

Cordwell's catalog blurb generator writes short product descriptions from a structured attribute table. It is Task B: the reference-free half of the day. There is no gold blurb for any product and there never will be, because nobody is going to hand-write 12,000 of them and then maintain them.

Item	Value
Catalog size	About 12,000 live SKUs
Blurbs generated per release	About 9,000. A seasonal release regenerates most of the catalog.
Release cadence	Roughly every six weeks, plus emergency patches
Where a blurb lands	The product page, the mobile app, and the printed seasonal circular
What you have	The attribute table, a style guide, and 40,000 historical blurbs written by the merchandising team over nine years
What you do not have	A single reference blurb for any current product, and no labeled judgement of whether any blurb is good
Judge budget	500 dollars of hosted judge calls per release
Human budget	4 hours of merchandising review per release
Team	Two engineers. The freeze starts in five weeks.

YOUR ASSIGNMENT Design the gate. Say what runs on what, what each layer costs, what it catches, and what gets through. Every line must survive arithmetic, and the totals have to fit inside 500 dollars and four hours.

4. The source of truth

One product, so the table is readable. In production there are 12,000 of these rows. This is the structured input the generator reads, and it is the only thing that can be called a fact about this product.

SKU	CW-4200
product_type	string trimmer
power	cordless
battery_volts	40
battery_amp_hours	2.5
battery_included	true
cutting_swath_in	15
line_diameter_in	0.080
weight_lb	8.4
runtime_min	45
shaft	straight
shaft_telescoping	false
edging_capable	true
attachment_compatible	false
water_resistance_rating	none
warranty_years	3

The style guide, abridged to what a check could test

#	Rule
1	Between 25 and 45 words.
2	Second person, present tense. Address the customer as you.
3	Lead with the primary benefit, not the model number.
4	No superlatives. No best, finest, ultimate, unbeatable.
5	No comparative claims against other brands.
6	State no more than two specifications numerically.
7	Close with the warranty term.
8	Never claim weather resistance, water resistance, or all-season use unless the attribute table rates it.

5. The slate

Eight candidate blurbs for the CW-4200, straight out of the generator. Some are shippable. Some are not. You will triage these in Round 1, so read them now but do not start sorting.

BLURB 1

Trim tight edges and finish borders in one pass. The 15-inch swath covers ground fast, and a single charge keeps you cutting for 45 minutes. Backed by a 3-year warranty.

BLURB 2

Clear overgrown borders without stopping. The 17-inch cutting swath handles wide runs quickly, and the included battery is ready out of the box. Backed by a 3-year warranty.

BLURB 3

Start with the trimmer head, then swap on the edger or cultivator attachment when the job changes. One powerhead, three tools. Backed by a 3-year warranty.

BLURB 4

Get through the whole yard on one charge. The battery runs 45 minutes, and a spare spool of line is included so you can reload without a trip to the store. Backed by a 3-year warranty.

BLURB 5

Head out early and keep working. This trimmer shrugs off morning dew and light rain, so a damp start never costs you the afternoon. Backed by a 3-year warranty.

BLURB 6

The best cordless trimmer in its class. At 8.4 pounds it is lighter than the leading brand, and it cuts a wider path than anything at this price. Backed by a 3-year warranty.

BLURB 7

Tackle your yard with confidence. This cordless string trimmer delivers reliable performance for homeowners who want quality results. The 15-inch swath makes quick work of overgrowth. Backed by a 3-year warranty.

BLURB 8

The Cordwell CW-4200 cordless string trimmer is designed for homeowners who need a dependable tool for regular yard maintenance, and it features a straight shaft configuration along with a 15-inch cutting swath that allows the operator to cover large areas of the property efficiently during routine trimming and edging tasks throughout the growing season.

BEFORE YOU START COUNTING WORDS Seven of these eight satisfy every mechanical style rule in Section 4. Length, warranty close, and specification count will separate exactly one blurb from the rest. If your gate is a word counter, you are about to publish six defects.

6. The price list

Rate sheet for Round 2. Costs are per release of 9,000 blurbs unless stated. You will be asked to total these, so treat the numbers as real.

Check	Build cost	Cost per release	What it can look at
Attribute grounding: extract numerics and categoricals from the blurb, compare against the table row	About 2 engineer-days	Free. Milliseconds per blurb.	Every claim whose value appears in the table
Mechanical style rules: word count, warranty close, banned-word list, specification count	About 1 engineer-day	Free	Anything expressible as a rule over the text
Near-duplicate check: embed the blurb, compare against the 40,000 historical blurbs	About 1.5 engineer-days	Free. Local embedding model, about 4 minutes for the batch.	How similar this blurb is to things already published
Judge, direct scoring, one JSON call returning several verdicts	About 3 engineer-days	\$0.038 per blurb	The blurb, the attribute row, and the style guide
Judge, pairwise against the incumbent blurb	About 1 engineer-day on top	\$0.038 per call	Two blurbs side by side
Human reviews one blurb against the table	None	About 40 seconds	Anything, slowly

Arithmetic you will need

Judge, direct, on all 9,000 $9000 \times 0.038 = \$342$
 Human review, 4 hours at 40 s each $14400 / 40 = 360 \text{ blurbs (4\% of the release)}$
 95% interval on a measured pass rate, roughly $\pm 1.96 \times \sqrt{p(1-p) / n}$:

n = 100	p = 0.95	-> +/- 4.3 points
n = 360	p = 0.95	-> +/- 2.3 points
n = 900	p = 0.95	-> +/- 1.4 points
n = 9,000	p = 0.95	-> +/- 0.5 points

Two of the rows on the price list are free and one of them is not obviously a check at all. Working out what the free rows can and cannot cover is most of Round 1.

7. Round 0: first allocation

Silent. Individual. Do not discuss.

Write how you would split the two budgets, before you know anything. Percentages are fine.

- The 500 dollars: what fraction goes to gating this release, and what fraction to anything else?
- The 4 human hours: what are those people actually doing?
- One thing you are assuming that you have not checked.

Almost everyone's first instinct is that the humans review blurbs. Hold on to whether you wrote that, because the second line is where this session usually changes somebody's mind.

8. Round 1: The Slate

Group. The clock matters.

Triage all eight blurbs against the attribute table and the style guide. Fill in Section 10, Part 1 as you go. Speed counts, but a false accusation costs you as much as a miss.

- 1.1** For each blurb: ship or block. Where you block, name the defect in a few words. One of the eight is genuinely clean; if you cannot find it, you are being too harsh, and a gate that blocks good blurbs has a cost that lands on the merchandising lead sitting next to you.
- 1.2** Sort your defects into kinds. You should end up with fewer kinds than defects. Two blurbs are wrong in ways that look different on the page and would be caught by exactly the same check; find them.
- 1.3** This is the question the rest of the session rests on. For each defect, decide which column it belongs in.

Column	Meaning
Free	A program with no understanding of English could catch it, given the attribute table and a list of rules
Not free	Catching it requires something that understands what the sentence is claiming

Count them. The ratio between those two columns is the single most important number in this discussion, and it decides how you spend the 500 dollars.

- 1.4 Two of the eight are defective in ways that no attribute check and no style rule will ever catch, and they fail for completely different reasons. Identify both and say what each one would need. One of them is not about facts at all, and it is the one your customers would notice first.
- 1.5 Blurb 4 claims something the attribute table neither confirms nor denies. Explain precisely why that is harder to check than blurb 2, which contradicts the table outright. Your answer should be about the shape of the data, not about how clever the check is.

9. Round 2: spend it

Group. Fill in Section 10, Part 2. Every line needs a number.

- 2.1 Build the gate. Order your layers by cost, cheapest first, and for each one state what it runs on, what it costs per release, and what fraction of the release it passes to the next layer. Your total must fit inside 500 dollars and four hours.
- 2.2 Somebody proposes pairwise comparison against the incumbent blurb, because slide 44 said it reaches higher agreement with humans. Price it honestly on all 9,000 and remember that the control for position bias is not optional. Then say whether it fits, and if it does not, what you would run it on instead.
- 2.3 You want to know whether this release is worse than the last one. Suppose the previous release passed 96 percent of blurbs and this one passes 93. Using the interval arithmetic in Section 6, roughly how many blurbs would you have to judge before you would believe a three-point drop is real rather than noise? Now price that number in dollars, and price the same number in human hours. Put the two figures next to each other and say what you notice.

This is the question that reorganizes the budget. Do not move on until both numbers are written down.
- 2.4 Allocate the four human hours and defend it. At 40 seconds each that is 360 blurbs, or four percent of the release. State exactly what those people produce, and what it is worth on the next release as well as this one.
- 2.5 You have 40,000 historical blurbs. Somebody proposes using them as references: for each new blurb, find the most similar historical blurb and score against it with ROUGE or chrF++. Say what that measures, what it does not, and what those 40,000 blurbs are genuinely worth instead. There is at least one use for them on your price list already.
- 2.6 Name what gets through your finished gate. Be specific enough that somebody could write the blurb that beats it. If you cannot, you have not finished designing.
- 2.7 The freeze is in five weeks and you have two engineers. Your price list has build costs on it. If you can only build two of the layers before the freeze, which two, and what do you do about the rest?

10. Your deliverable: the Budget Sheet

Part 1. The slate

Blurb	Ship or block	The defect	The kind	Free or not free
1				
2				

Blurb	Ship or block	The defect	The kind	Free or not free
3				
4				
5				
6				
7				
8				

Free defects: _____ Not-free defects: _____

Part 2. The gate

Layer	Runs on what, and how many	Cost per release	What it catches, and what it passes on
1			
2			
3			
4			

Total dollars: _____ of 500 Total human hours: _____ of 4 What the humans produce:

What gets through: _____

11. Round 4: report out

#	State this	What makes it count
1	Your budget line: dollars spent, hours spent, and what each bought	Two numbers and two purposes. If the humans are reviewing blurbs, say so and defend it.
2	The ratio you found in Round 1, free defects against not-free defects	This is the number that justifies everything else on your sheet.
3	The thing you refuse to spend money on, and why	Name the check and the reason. "Too expensive" is not a reason on its own.

#	State this	What makes it count
4	What gets through your gate	Specific enough that another group could write it.

Statement 1 is where the room will disagree most, and the disagreement is worth more than the answer. There is more than one defensible allocation, but there are several indefensible ones and they are common.

12. If your group is stuck

Three levels per question. Take one at a time and stop as soon as you are moving again. None of these contains an answer.

Question 1.3 | Free or not free

Level 1. Ask one question of each defect: is there a field in the attribute table that settles it? If yes, a program can settle it too.

Level 2. The table is structured data, so any claim about a value in it reduces to a comparison. Numbers, true and false flags, and named categories are all settleable this way. Ask which of your defects are about something the table has a row for.

Level 3. Structured sources make a surprising share of factual checking free, because the check is a lookup rather than an act of comprehension. That is the difference between this task and a summarization task where the source is prose. Count how many of your eight fall on the free side before you decide the judge is the centre of your design.

Question 1.5 | Why an unsupported claim is harder than a contradiction

Level 1. One blurb says a number that disagrees with the table. Another says something the table simply has no row about. Compare what a program has to do in each case.

Level 2. To catch a contradiction you compare two values that both exist. To catch an unsupported claim you have to notice that a claim was made at all, and then establish that nothing anywhere supports it.

Level 3. Contradiction is a closed-world problem: the answer is present and you check against it. Unsupported addition is an open-world problem: absence of evidence has to be recognized as such, which means something must first identify what claims the sentence is making. That identification step is what you are paying the judge for, and it is why this residue is where the money goes.

Question 2.2 | Pricing pairwise

Level 1. Work out the cost of one pairwise comparison before you work out the cost of nine thousand.

Level 2. Slide 45 named a control for position bias and it is not optional. Read what the control requires, then count how many judge calls one comparison actually costs.

Level 3. Running every pair twice with the order swapped doubles the call count, so pairwise on the full release is two calls per blurb, not one. Multiply that out against the budget and you will find it does not fit. That does not mean pairwise is wrong; it means pairwise is something you run on a sample to answer a different question than the gate is answering.

Question 2.3 | How many do you have to judge

Level 1. Use the interval table in Section 6. A three-point difference is only believable if your interval is comfortably smaller than three points.

Level 2. Look at what n buys you: 360 gives you plus or minus 2.3 points, and you are trying to see a three-point move in two separate measurements, each with its own interval. Work out roughly what n gets both intervals small enough to separate.

Level 3. You need something close to a thousand judged items per release to see a three-point move with any confidence. Now price a thousand items two ways, once at the judge's rate and once at 40 seconds of a person, and put the two numbers side by side. One of them fits in your budget with room to spare and the other exceeds it by more than a factor of two. That comparison should change what you think the humans are for.

Question 2.4 | What the humans are actually for

Level 1. Whatever the humans do, they do it to 360 items out of 9,000. Ask what those 360 are worth if you throw them away after this release.

Level 2. Every automated layer you have proposed makes a claim about quality. Ask which of those layers you have any evidence for, and what evidence would look like.

Level 3. A judge is an instrument and an instrument you have not compared against human judgement reports numbers of unknown meaning. Four hours of human labeling produces something that lasts: a measured agreement between your judge and your people, which makes every judged blurb in every future release interpretable. Four hours of human blurb-reviewing produces 360 reviewed blurbs and nothing else. Both are defensible. Only one of them compounds.

13. Vocabulary for this discussion

Term	Working definition in this context
Closed-world claim	A claim about a value the source has a field for. Settleable by lookup, which makes it nearly free to check when the source is structured.
Open-world claim	A claim the source neither confirms nor denies. Catching it requires identifying that a claim was made, which is the expensive step.
Gate versus monitor	A gate blocks bad output before it ships and must run on everything. A monitor detects that a release went wrong and can run on a sample. Same check, different job, wildly different cost.
Calibration	Measuring how often your judge agrees with your people. Without it, the judge's pass rate is a number with no known relationship to quality.
Statistical power	Whether your sample is large enough to see the difference you care about. Power is bought with sample size, and sample size has a price per item.
Compounding spend	Work that keeps paying on later releases. A calibrated judge compounds; a reviewed blurb does not.
Block rate	The fraction of blurbs your gate refuses to publish. Every point of it lands on somebody who has to unblock a product page.
Template collapse	Output that is accurate, in style, and indistinguishable from what you already published thousands of times. No factual check will ever flag it.